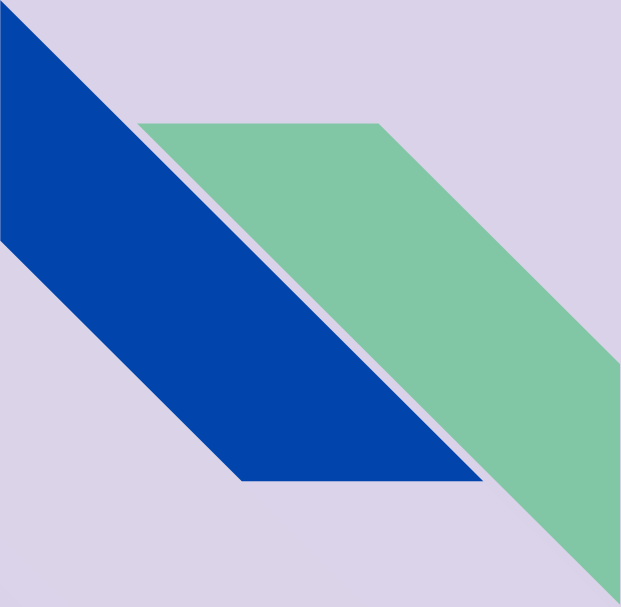




Hospital Stay Duration Analysis

UC Berkeley ML and AI Certification 2025

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Overview

This project analyzes a comprehensive hospital dataset from New York State to understand and predict the length of patient hospital stays. It involves exploratory data analysis, feature engineering, and predictive modeling using regression techniques.

Dataset Description: The dataset contains records of hospital stays including patient demographics, admission details, clinical severity scores, financial charges, and outcomes.



Project Goals

Hospital Length of Stay (LOS) is a critical indicator of healthcare efficiency, resource utilization, and patient outcomes. Accurately predicting how long a patient will remain hospitalized allows healthcare providers to optimize bed management, staff allocation, and discharge planning, ultimately reducing overcrowding and improving patient care.

However, LOS is influenced by a complex combination of factors — including a patient's age, severity of illness, admission type, comorbidities, and discharge disposition — many of which interact in nonlinear and unpredictable ways. Hospitals often rely on retrospective averages or clinician judgment, which can be imprecise and lead to resource bottlenecks or delayed discharges.



Project Goal Continued

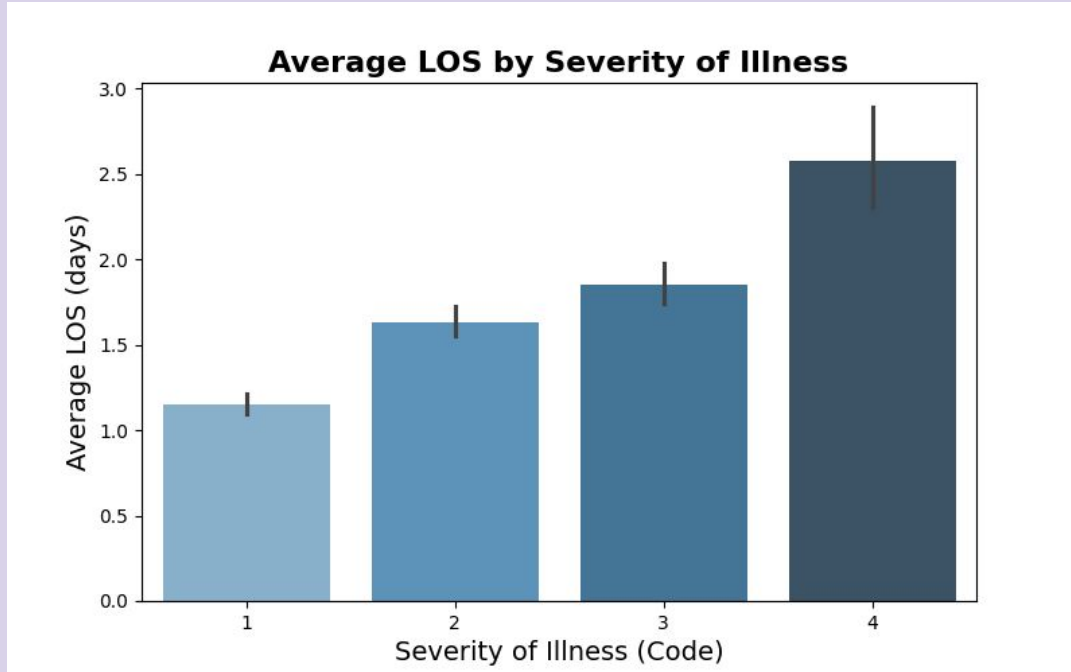
This project aims to address that gap by developing a **machine learning–based LOS prediction model** capable of identifying patterns in patient characteristics and administrative data that correlate with prolonged hospital stays. By leveraging data-driven predictions, hospitals can:

- Forecast demand for inpatient beds and critical care resources,
- Identify patients at risk of extended stays early in their admission, and
- Enhance care coordination between clinical and administrative teams.

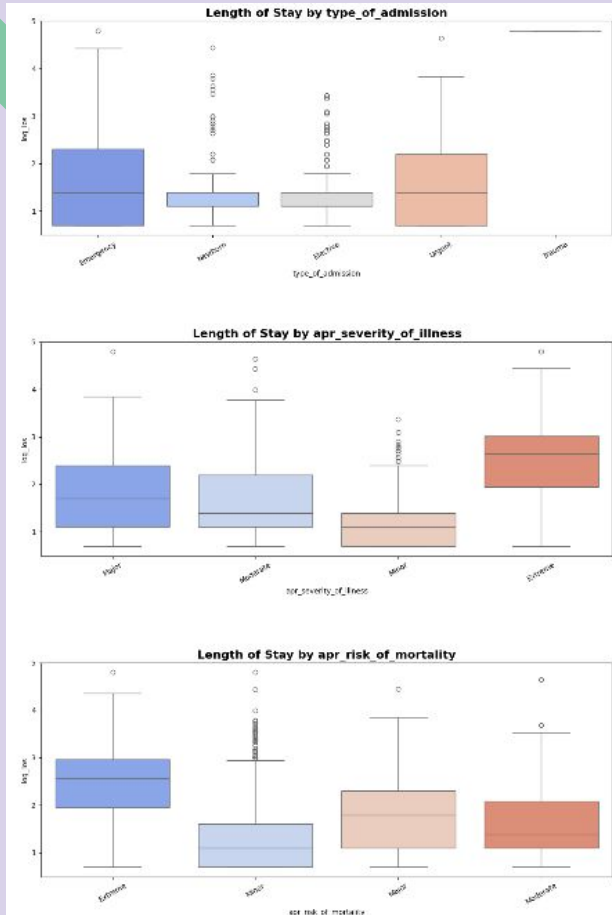
The ultimate goal is to use predictive modeling to support **data-informed decision-making** in hospital operations — improving both patient outcomes and system efficiency.

Multivariate Numerical Analysis

Severity of Illness code -> the higher the code number the longer the length of stay



Multivariate Categorical Analysis



Type of Admission: Emergency cases have the longest average stays, while elective and newborn admissions are generally shorter.

Severity of Illness: Length of stay increases consistently with severity level, making it one of the strongest predictors of LOS.

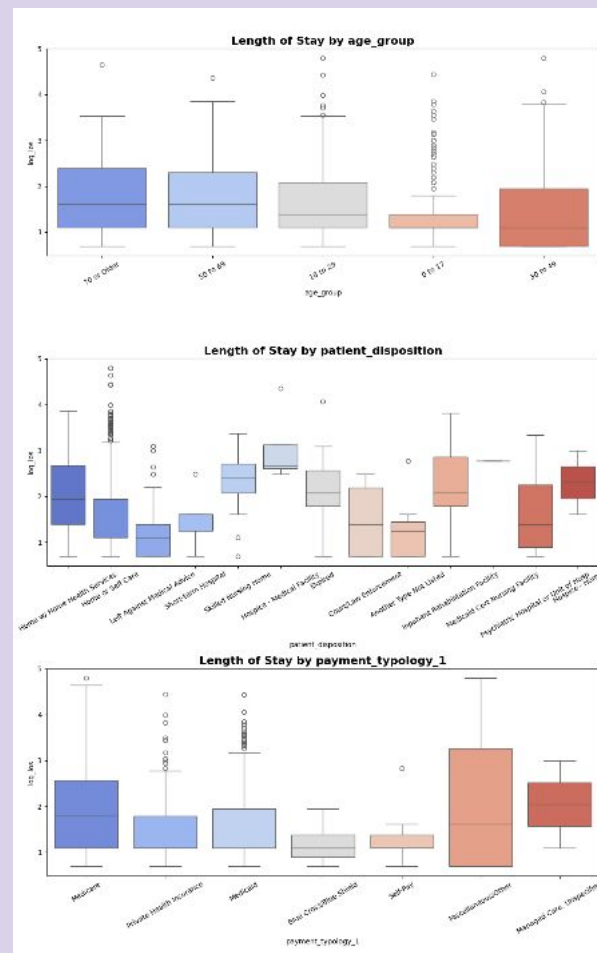
Risk of Mortality: Higher mortality risk corresponds to longer stays, with extreme-risk patients showing the widest variability.

Multivariate Categorical Analysis Cont.

Age Group: Older patients, especially those aged 70+, stay significantly longer than younger age groups.

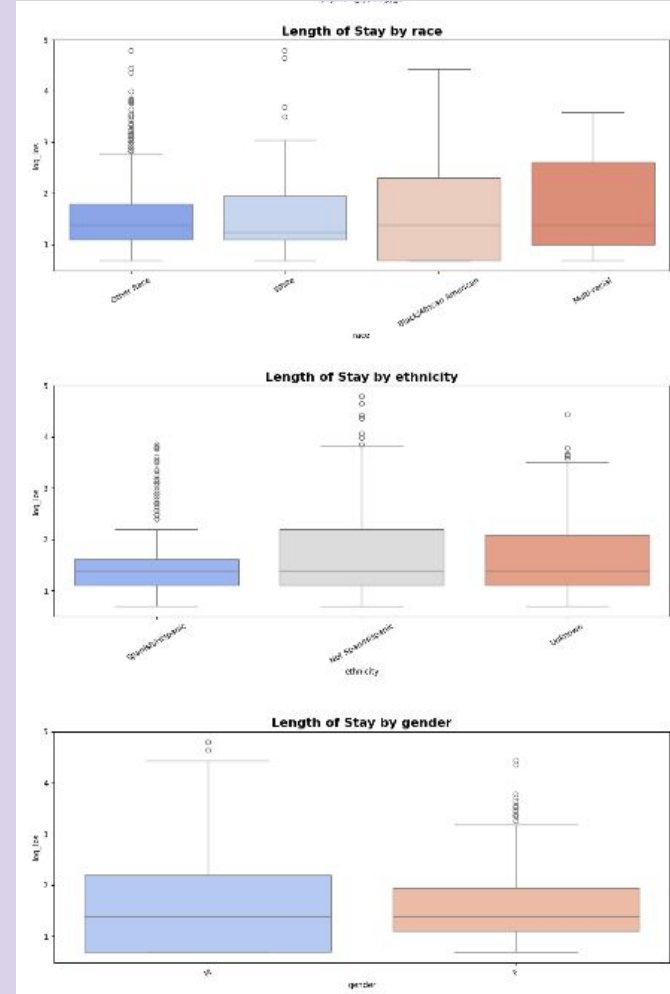
Patient Disposition: Patients discharged home have short stays, while transfers to rehab or skilled nursing show the longest.

Payment Typology: Medicare patients stay longest on average, reflecting higher age and condition complexity.

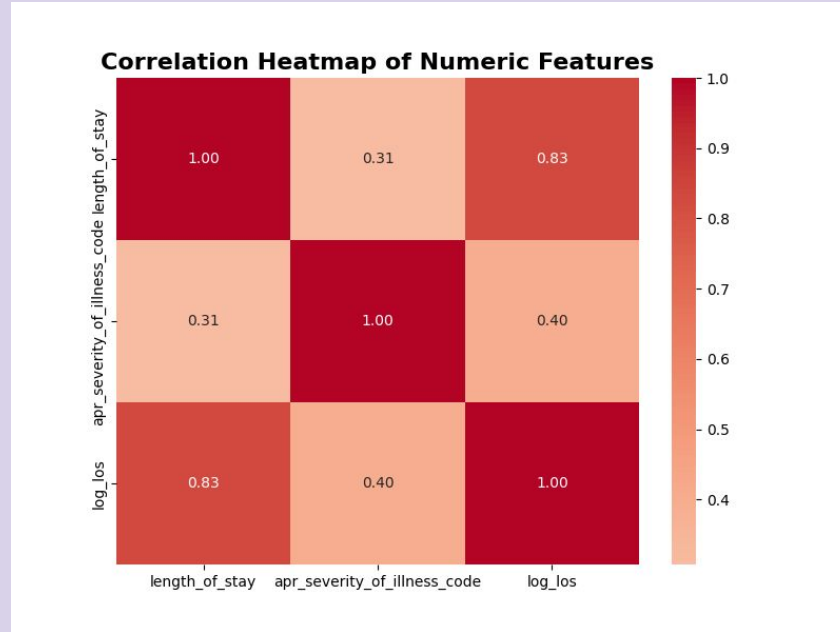


Multivariate Categorical Analysis Cont.

- **Gender:** LOS distribution is similar for males and females, showing no meaningful gender difference.
- **Race:** No major differences in LOS are observed across racial groups, though smaller sample sizes limit reliability.
- **Ethnicity:** LOS is similar across Hispanic and non-Hispanic groups, but missing data in the “Unknown” category adds uncertainty.



Correlation Map



A moderate positive correlation ($r = 0.40$) between LOS and illness severity suggests that more severe cases generally lead to longer hospitalizations.

Model Comparison

	Model	CV_RMSE	Test_RMSE	Test_MAE	Test_R2	Best_Params
0	Lasso	0.773122	0.690103	0.540167	0.225621	{'model__alpha': 0.001}
1	Ridge	0.769427	0.690266	0.542261	0.225256	{'model__alpha': 10.0}
2	ElasticNet	0.769445	0.690536	0.542746	0.224648	{'model__alpha': 0.012, 'model__l1_ratio': 0.05}
3	LinearRegression	0.774882	0.691093	0.540444	0.223397	{}
4	SVR	0.770595	0.696711	0.524785	0.210720	{'model__C': 1.0, 'model__epsilon': 0.2, 'mode...
5	GradientBoosting	0.770375	0.703971	0.549803	0.194185	{'model__learning_rate': 0.1, 'model__max_dept...
6	RandomForest	0.770507	0.708085	0.549758	0.184738	{'model__max_depth': 10, 'model__min_samples_s...
7	DecisionTree	0.794090	0.733491	0.575606	0.125187	{'model__max_depth': 5, 'model__min_samples_sp...
8	KNN	0.782289	0.737468	0.572648	0.115674	{'model__n_neighbors': 11, 'model__weights': '...
9	Dummy	0.849627	0.784490	0.653629	-0.000693	{}

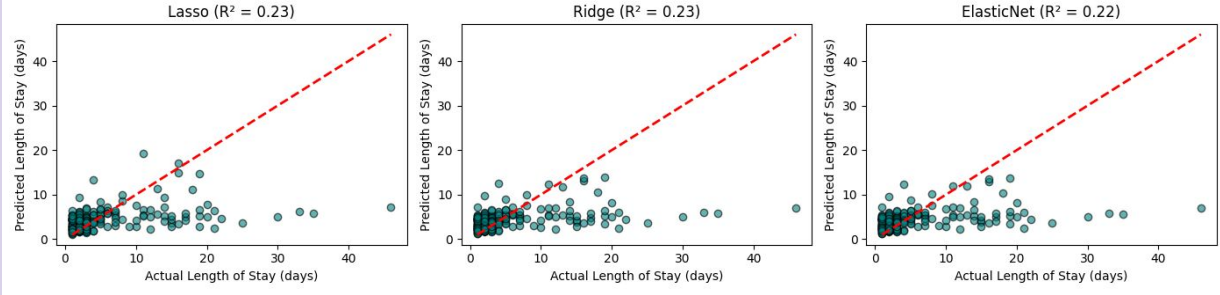
Interpretation:

Regularized linear models (Ridge, Lasso, ElasticNet) perform best, suggesting the relationship between features (age group, severity, risk, etc.) and log(LOS) is mostly linear with limited benefit from non-linear or tree-based methods.

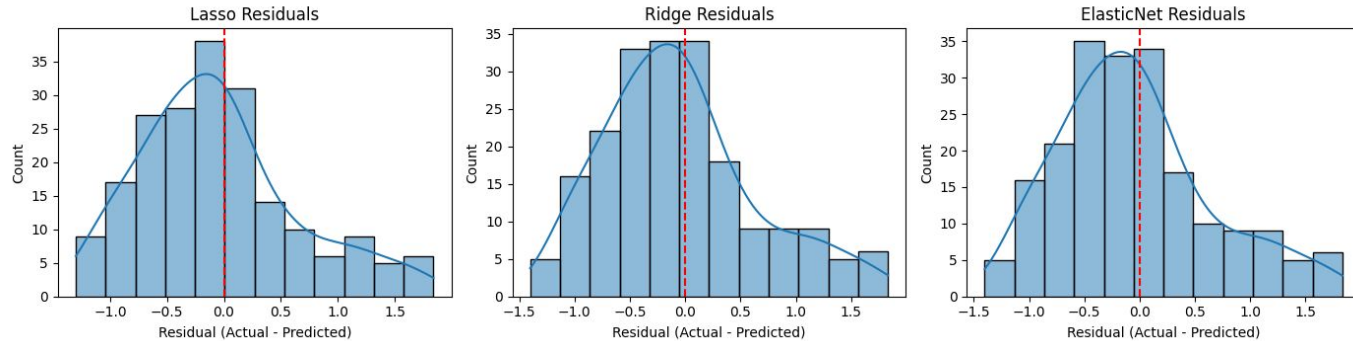
Lasso Vs. Ridge Vs. ElasticNet

Overall, the three models (Lasso, Ridge, ElasticNet) perform nearly identically, confirming the relationship between predictors and LOS is largely linear. Ridge regression displays the most balanced residual spread, supporting its selection as the most stable and well-calibrated model

Predicted vs. Actual Hospital Length of Stay (Back-Transformed)



Residual Distributions — Top Linear Models





Improving Ridge Model

Tried to improve the model by using the best value of alpha. -> Worked by didn't help that much with improving the model. Why?

- Ridge tuning only adjusts how much to shrink coefficients, not the form of the relationship. model already found the sweet spot between bias and variance — the flat region in the curve shows that performance has plateaued.
- Tuning worked but it just confirmed stabilization and not improvement



Model Evaluation Conclusion

Regularized linear models (Lasso and Ridge) provided the best balance of accuracy, stability, and interpretability. While Gradient Boosting and Random Forest captured some complexity, they did not significantly outperform linear models due to the dataset's categorical structure.

The results demonstrate the feasibility of machine learning–based LOS prediction for improving hospital capacity management, discharge coordination, and overall operational efficiency.

Among them, the Ridge Regression model proved to be the most stable, showing consistent residual patterns and reliable performance across folds, making it a strong candidate for practical hospital implementation.

Feature Selection

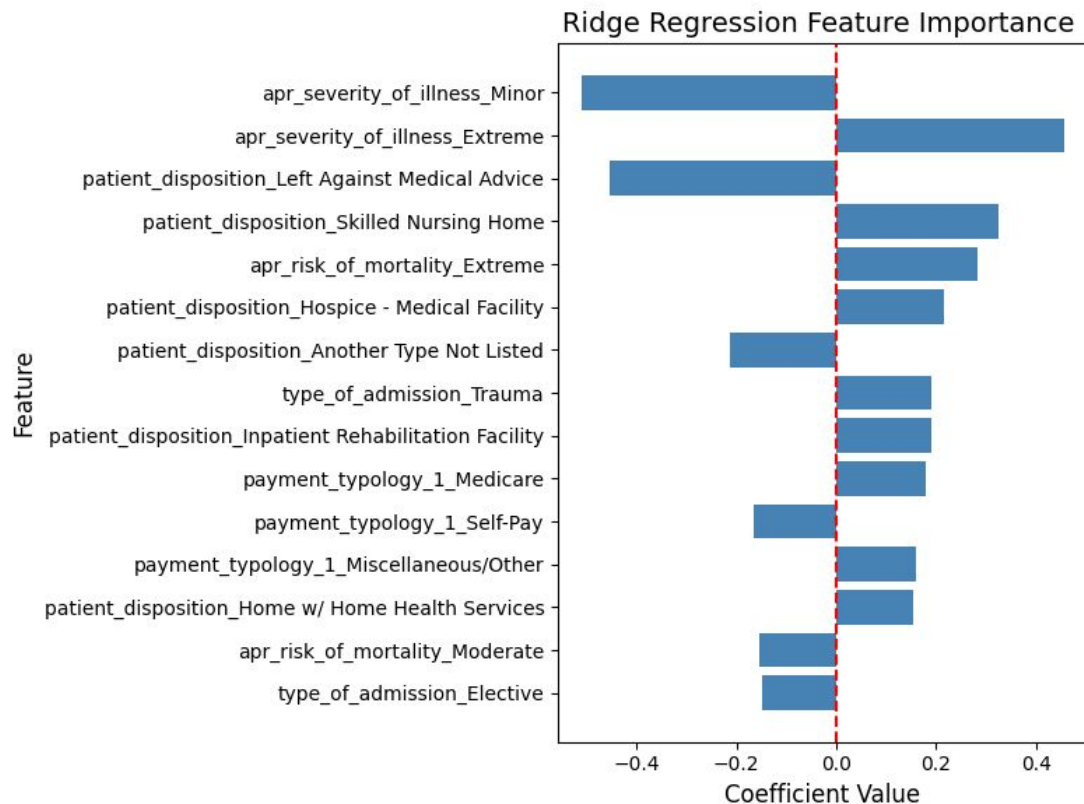
Summary

Features that increase LOS

- Higher severity and mortality risk
- Complex discharge destinations
- Trauma admissions

Features that decrease LOS

- Minor severity and AMA discharges
- Expired or self-pay patients (shorter stays)
- Certain demographics and non-complex cases





Summary

This project predicted hospital length of stay (LOS) using demographic, admission, and clinical severity data to improve hospital efficiency.

Feature selection showed that severity of illness, patient disposition, risk of mortality, and admission type were the strongest predictors of LOS.

Among tested models, Ridge and Lasso regression achieved the best performance ($R^2 \approx 0.22$, RMSE ≈ 0.69), with Ridge proving the most stable across folds.

The model demonstrates that machine learning can support capacity planning, discharge coordination, and resource optimization in hospitals.



Business Recommendations

Business Recommendations

- 1. Prioritize high-severity and emergency admissions for early discharge planning.**
Since *severity of illness* and *admission type* are the strongest predictors of longer stays, hospitals should flag these patients at admission for proactive case management and resource allocation.
- 2. Enhance capacity and staffing forecasts based on predicted LOS.**
Predictive LOS models can support scheduling of staff, beds, and ICU resources—especially in high-volume emergency departments—to reduce bottlenecks and boarding times.
- 3. Use LOS predictions to optimize post-acute care coordination.**
Patients likely to require skilled nursing or rehab transfers can be identified earlier, enabling smoother handoffs and reducing discharge delays.
- 4. Monitor and reduce variability among high-risk groups.**
Outliers (e.g., extreme severity or mortality risk patients) contribute disproportionately to bed occupancy; tracking these cases can improve throughput and cost efficiency.
- 5. Incorporate predictive insights into administrative dashboards.**
Integrating LOS forecasts into hospital management systems allows real-time decision-making for admissions, transfers, and discharges.



Technical Recommendations

Expand feature set with richer clinical and operational data.

Include continuous variables such as *exact age*, *lab results*, *comorbidity scores*, *ICU hours*, and *procedure counts* to capture nonlinear patterns missed by the current categorical dataset.

Test advanced ensemble models.

Explore **XGBoost**, **LightGBM**, or **Stacking Regressors** (e.g., Ridge + Gradient Boosting) to leverage nonlinear interactions and potentially improve R^2 beyond 0.30.

Implement explainability techniques.

Use **SHAP** or **LIME** for model interpretation, helping clinicians and administrators understand how each feature drives predicted LOS.

Deploy and validate model in production.

Develop a simple web-based dashboard or API to integrate predictions into EMR or hospital workflow, followed by continuous model retraining as new data arrive.